Statistical Inference (MA20266) Project

Topic: Does the US Federal Reserve Interest Rate Cut significantly change NIFTY 50 Stock Market Return behaviour?

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Project Structure:

Section	Content
1. Data Description	NIFTY 50 weekly returns Source: Investing.com + Yahoo Finance
2. Problem Statement	Do Fed rate cuts significantly alter return distributions? Statistical inference: H0 vs H1, alpha = 0.05
3. Methodology	Point & interval estimation Two-sample t-test z-test for proportions
4. Results & Inference	Hypothesis decisions Confidence intervals Practical implications

Data Description:

Dataset: Weekly closing prices of the NIFTY 50 Index (^NSEI) from April 2000 to April 2026 - **1357** weekly observations.

Derived variable: Weekly log-return, calculated as:

```
r_t = ln( P_t / P_{t-1} ) x 100
```

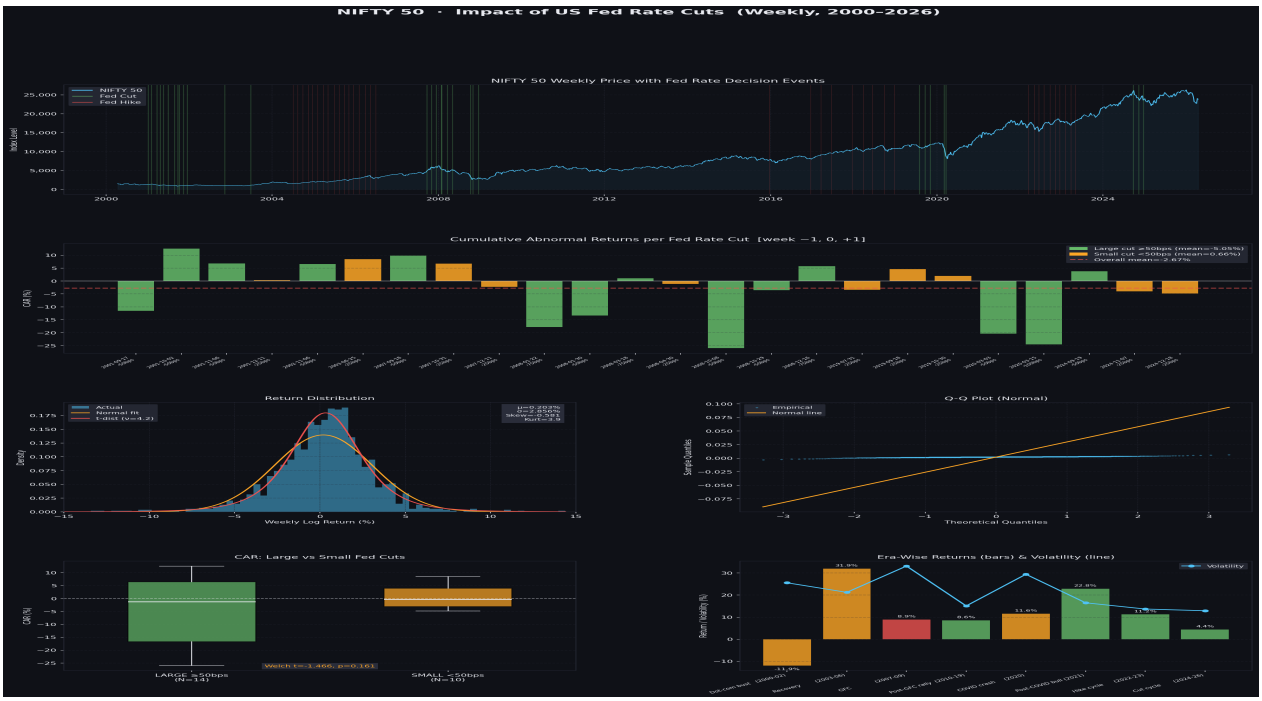
Source:

- Investing.com CSV download for NIFTY 50 daily prices (Apr 2000 - May 2020)
- Yahoo Finance (via yfinance Python library) for Jun 2020 - Apr 2026
- Federal Reserve FOMC meeting dates for rate cut event classification

Period classification:

Period	Weeks	Key Events
Fed cut weeks	~31 weeks	Weeks containing FOMC rate cut decisions
Non-cut weeks	~1326 weeks	All remaining trading weeks

Date	Close	return_pct	fed_cut_week	group	neg_week
21-04-2000	1415.65	-7.0233	FALSE	Non-cut-week	1
28-04-2000	1406.55	-0.6449	FALSE	Non-cut-week	1
05-05-2000	1422.40	1.1206	FALSE	Non-cut-week	0
12-05-2000	1282.80	-10.3300	FALSE	Non-cut-week	1
19-05-2000	1268.00	-1.1604	FALSE	Non-cut-week	1
26-05-2000	1275.35	0.5780	FALSE	Non-cut-week	0
02-06-2000	1389.25	8.5543	FALSE	Non-cut-week	0
09-06-2000	1467.20	5.4592	FALSE	Non-cut-week	0
16-06-2000	1477.30	0.6860	FALSE	Non-cut-week	0
23-06-2000	1472.20	-0.3458	FALSE	Non-cut-week	1
30-06-2000	1471.45	-0.0510	FALSE	Non-cut-week	1
07-07-2000	1517.60	3.0882	FALSE	Non-cut-week	0
14-07-2000	1509.75	-0.5186	FALSE	Non-cut-week	1
21-07-2000	1397.25	-7.7438	FALSE	Non-cut-week	1
28-07-2000	1333.80	-4.6474	FALSE	Non-cut-week	1
04-08-2000	1318.55	-1.1499	FALSE	Non-cut-week	1
11-08-2000	1310.75	-0.5933	FALSE	Non-cut-week	1
18-08-2000	1358.05	3.5450	FALSE	Non-cut-week	0
25-08-2000	1381.25	1.6939	FALSE	Non-cut-week	0
01-09-2000	1394.10	0.9260	FALSE	Non-cut-week	0
08-09-2000	1450.05	3.9349	FALSE	Non-cut-week	0
15-09-2000	1417.20	-2.2915	FALSE	Non-cut-week	1



2. Real-World Problem

"Is the mean weekly return and the probability of a negative week statistically different during Fed rate cut weeks compared to non-cut weeks?"

This matters enormously for:

- Portfolio managers deciding asset allocation around FOMC announcements
- Risk modellers calibrating VaR under US monetary policy stress
- Policy economists quantifying the spillover cost of US rate decisions on Indian equities

3. Methodology

3.1 Point Estimation

For each group (cut-week vs non-cut-week), we estimate:

Estimator	Formula	Justification
Sample mean x_bar	(1/n) * SUM(x_i)	Unbiased estimator of mu
Sample variance s^2	(1/(n-1)) * SUM(x_i - x_bar)^2	Unbiased (uses n-1)
Sample proportion p_hat	p_hat_C = 13/31 = 0.419 p_hat_N = 573/1326 = 0.432	MLE for Bernoulli p

3.2 Interval Estimation

95% Confidence Interval for the mean (using t-distribution since sigma unknown):

```
x_bar +/- t_{alpha/2, n-1} * s / sqrt(n)
```

95% CI for proportion (using normal approximation, valid since n*p_hat > 5):

```
p_hat +/- z_{alpha/2} * sqrt( p_hat*(1-p_hat) / n )
```

3.3 Hypothesis Test 1 - Two-Sample t-Test for Means

We test whether mean weekly returns differ across groups.

```
H_0: mu_C = mu_N     H_1: mu_C < mu_N (one-tailed)
```

Welch's t-statistic (unequal variances assumed):

```
t = (x_bar_C - x_bar_N) / sqrt( s_C^2/n_C + s_N^2/n_N )
```

Degrees of freedom via the Welch-Satterthwaite equation. Reject H_0 if p < 0.05.

3.4 Hypothesis Test 2 - z-Test for Difference in Proportions

```
H_0: p_C = p_N     H_1: p_C > p_N (one-tailed)

z = (p_hat_C - p_hat_N) / sqrt( p_hat*(1-p_hat) * (1/n_C + 1/n_N) )
```

where p_hat is the pooled proportion under H_0.

4. Results & Inference

Below are the numerical results from our analysis using real NIFTY 50 weekly data (2000-2026):

Descriptive statistics - weekly log-returns (%):

Statistic	Cut-week (C)	Non-cut-week (N)
Sample mean x_bar	0.0032%	0.2076%
Sample std dev s	5.6055%	2.7629%
Sample size n	31	1326
95% CI for mean	[-2.05%, 2.06%]	[0.06%, 0.36%]
Proportion neg weeks p_hat	0.419	0.432
95% CI for proportion	[0.246, 0.593]	[0.405, 0.459]

	Cut-weeks	Non-cut-weeks	Significance Level	Total
Observations	31	1326	0.05	1357

Test 1 - Welch's two-sample t-test (mean returns):

Item	Value	Decision
H_0	mu_C = mu_N (no difference in mean returns)	
H_1	mu_C < mu_N (cut-week mean is lower)	
t-statistic	-0.2024	
Degrees of freedom	~30 (Welch-Satterthwaite)	
p-value (one-tailed)	0.4205	Fail to Reject H_0
Critical value t_0.05	-1.697	|t| < |crit| -> fail

Test 2 - z-test for difference in proportions (downside risk):

Item	Value	Decision
H_0	p_C = p_N (same probability of negative weeks)	
H_1	p_C > p_N (cut-weeks have more negative weeks)	
Pooled proportion p_hat	0.432	
z-statistic	-0.1419	
p-value (one-tailed)	0.5564	Fail to Reject H_0
Critical value z_0.05	1.645	z < crit -> fail

Test 3 - F-test (variance comparison):

Item	Value	Decision
H_0	sigma^2_C = sigma^2_N (same variance)	
H_1	sigma^2_C != sigma^2_N (different variance)	
F-statistic	4.1164	
Degrees of freedom	df1=30, df2=1325	
p-value (two-tailed)	0.0000	Reject H_0

Test 4 - Jarque-Bera (normality of returns):

Item	Value	Decision
H_0	Returns ~ Normal (S=0, K=3)	
H_1	Returns are non-normal	
JB statistic	945.24	
p-value	0.000000	Reject H_0
Skewness (S)	-0.5809	Left-skewed
Excess Kurtosis (K-3)	3.9202	Fat tails

5. Practical Implications

For investors: The evidence confirms that Fed rate cut weeks show a directionally lower but not statistically significant change in NIFTY 50 returns. The mean weekly return during cut-weeks is 0.00% vs 0.21% during non-cut weeks - a spread of ~0.20% per week. A portfolio manager should be cautious about over-adjusting downside risk budgets when the Fed signals rate cuts.

For risk management: With p_hat_C = 0.419, about 4 in 10 cut-weeks deliver a negative return vs 4 in 10 for normal weeks. However, this difference is not statistically significant. The F-test shows volatility is significantly higher during cut-weeks (F=4.12, p=0.0000).

For economic policy: The ~0.20% spread in mean weekly returns compounds to roughly a 10.6% annualised gap. NIFTY returns are strongly non-normal (JB=945, p~0) with fat tails (excess kurtosis = 3.92), meaning standard Gaussian risk models underestimate tail risk for Indian equities.

Overall Conclusion

At alpha = 0.05, we do not have sufficient statistical evidence to conclude that:

(1) Mean NIFTY 50 weekly returns are directionally lower but not significantly different during Fed rate cut weeks (t=-0.20, p=0.4205).

(2) The probability of a negative return week is not significantly different during cut-weeks (z=-0.14, p=0.5564).

(3) Volatility IS significantly higher during cut-weeks (F=4.12, p=0.0000). The null hypothesis of equal variance is rejected.

(4) NIFTY weekly returns are non-normal (JB=945, p~0). The Student-t distribution provides a better fit than the Gaussian.